

Regularization, Data Augmentation and Self-Supervised Learning

Efficient Deep Learning - Session 2



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2023

Sessions

- 1 Intro Deep Learning,
- 2 Data Augmentation and Self Supervised Learning,
- 3 Quantization,
- 4 Pruning,
- 5 Factorization,
- 6 Distillation,
- 7 Embedded Software and Hardware for DL,
- 8 Presentations for challenge.

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Why this session ?

Regularization

Constrain the training for faster convergence and better generalization.

Data Augmentation (DA)

Help generalization by sampling training examples from a larger distribution using randomized transforms.

Self-supervised Learning (SSL)

Exploit DA and regularization tricks for learning representations, without labels

Significance

- In some (most?) cases, DA regularizes training and is needed.
- Large networks can't be trained without regularization.

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Weight Decay

An old idea (Krogh and Herz 1991): ℓ_2 penatly term is added to the loss, limits the growth of model weights.

Has been shown to increase generalization and suppresses irrelevant model weights.

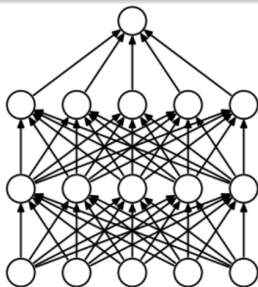
Ressources :

- <https://proceedings.neurips.cc/paper/1991/file/8eefcfd5990e441f0fb6f3fad709e21-Paper.pdf>
- https://ja.d2l.ai/chapter_deep-learning-basics/weight-decay.html
- Readily available in pytorch (optimizer options)

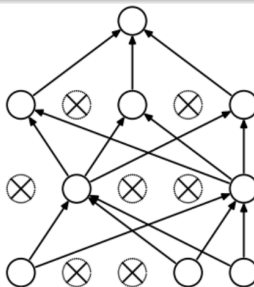
Regularization

Dropout

Randomly "drops" some units during training with a certain probability.



(a) Standard Neural Net



(b) After applying dropout.

- Was introduced to train very large networks
- Can prevent overfitting
- Adds hyperparameters : where to drop ? How often ?

<https://www.jmlr.org/papers/volume15/srivastava14a/srivastava14a.pdf>

Batch Normalization (Ioffe & Szegedy, 2015)

Normalize feature distributions to the standard distribution by learning batch statistics.

- Consider a batch X
- Calculate $m = E(X)$ and $\sigma = \text{Var}(X)$
- Compute $\hat{X} = \frac{X-m}{\sigma} * \gamma + \beta$
- m and σ are continuously updated across batches using running statistics, and γ and β are learnable parameters (by default set to 1 and 0, respectively)

Notes

- Has been shown to accelerate training, increase generalization
- Can remove the need for DropOut
- Should be included by default after convolutions

Data Augmentation using image transformations

Translations, rotations, Scaling, Shifting in RGB, Crops, ...



Image from Albumentations https://albumentations.ai/docs/examples/pytorch_classification/

Mixup, Cutout and Cutmix

Mixup

For a network F trained using Cross Entropy (CE),

- Sample x_i, x_j from the training data, associated to labels y_i, y_j .
- Defined mixed up data samples as $\tilde{x} = \lambda x_i + (1 - \lambda)x_j$
- $loss = \lambda CE(F(\tilde{x}), y_i) + (1 - \lambda)CE(F(\tilde{x}), y_j)$, where $\lambda \in [0, 1]$
- Train with backprop

Notes

- Has been shown to regularize training and achieves better generalization.
- Should be included most of the time when training classification networks !
- See Lab2.md for a proposed implementation

<https://arxiv.org/pdf/1710.09412.pdf>

Mixup, Cutout and Cutmix

	ResNet-50	Mixup [47]	Cutout [3]	CutMix
Image				
Label	Dog 1.0	Dog 0.5 Cat 0.5	Dog 1.0	Dog 0.6 Cat 0.4
ImageNet Cls (%)	76.3 (+0.0)	77.4 (+1.1)	77.1 (+0.8)	78.6 (+2.3)
ImageNet Loc (%)	46.3 (+0.0)	45.8 (-0.5)	46.7 (+0.4)	47.3 (+1.0)
Pascal VOC Det (mAP)	75.6 (+0.0)	73.9 (-1.7)	75.1 (-0.5)	76.7 (+1.1)

Table 1: Overview of the results of Mixup, Cutout, and our CutMix on ImageNet classification, ImageNet localization, and Pascal VOC 07 detection (transfer learning with SSD [23] finetuning) tasks. Note that CutMix significantly improves the performance on various tasks.

https://openaccess.thecvf.com/content_ICCV_2019/papers/Yun_CutMix_Regularization_Strategy_to_Train_Strong_Classifiers_With_Localizable_Features_ICCV_2019_paper.pdf

Self Supervised Learning

The principle

Learning useful representations from data without relying on labels. The model *creates* the labels based on the structure of the data through *pretext tasks*

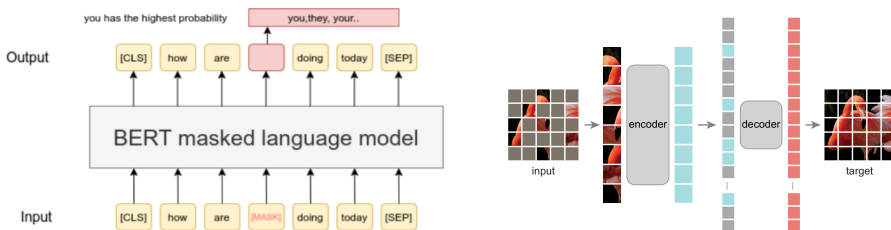
Why it is important

- **Scalability:** Since unlabeled data is much easier to obtain, models can scale to larger datasets and more diverse inputs
- **Robust Representations → Transferability:** It often leads to more generalizable representations that perform well across multiple tasks compared to models trained on a more limited label space

Self Supervised Learning

Types of SSL approaches

- **Masked Input Modeling:** Predicting missing part of the input



BERT: <https://arxiv.org/pdf/1810.04805>

Masked Autoencoders: <https://arxiv.org/pdf/2111.06377>

Self Supervised Learning

Types of SSL approaches

- **Masked Input Modeling:** Predicting missing part of the input
- **Contrastive Learning:** Pulling together similar representations and pushing apart dissimilar ones

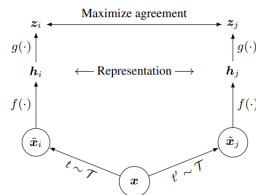
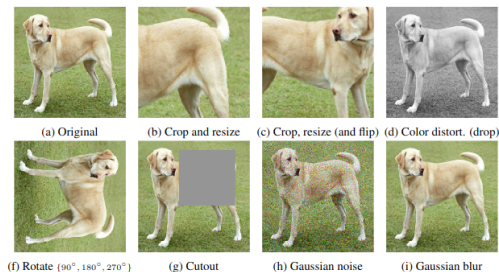
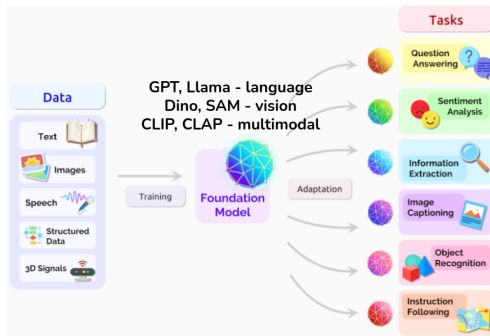


Figure 2. A simple framework for contrastive learning of visual representations. Two separate data augmentation operators are sampled from the same family of augmentations ($t \sim \mathcal{T}$ and $t' \sim \mathcal{T}$) and applied to each data example to obtain two correlated views. A base encoder network $f(\cdot)$ and a projection head $g(\cdot)$ are trained to maximize agreement using a contrastive loss. After training is completed, we throw away the projection head $g(\cdot)$ and use encoder $f(\cdot)$ and representation h for downstream tasks.

SimCLR: <https://arxiv.org/pdf/2002.05709>, BYOL: <https://arxiv.org/pdf/2006.07733>

Self Supervised Learning

SSL to pretrain foundation models

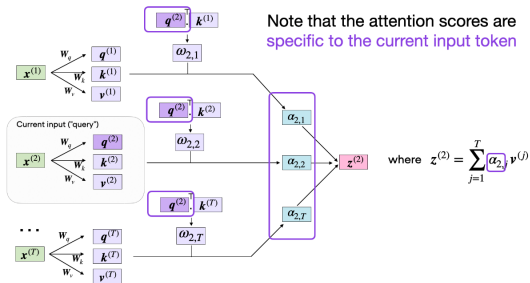


<https://arxiv.org/pdf/2108.07258.pdf>

Foundation Model

- Pretrained on internet-scale data with SSL
- Able to learn general features from data
- Perform (or can be easily adapted to) multipurpose tasks

Self-Attention



Self-Attention in foundation models

- Grasp relationships between parts of the inputs (context)
- Attention weights ω : dot product input query $\mathbf{q}$ and all other inputs key $\mathbf{k}$
- The input $\mathbf{x}$ is transformed in the context vector $\mathbf{z}$, which is an attention-weighted version of the original query input

<https://sebastianraschka.com/blog/2023/self-attention-from-scratch.html>

Self-Attention and Transformers

- Self-Attention is found in the basic architecture of foundation models: **Transformers**
- No convolutions, inputs are transformed taking in account attention weights
- Best generalization in many domains, but need large scale data

<https://arxiv.org/pdf/1706.03762.pdf>

